SOP – EFORT Robot Operation Procedure

1. SOP Details

Machine name : Effort Industrial Robot System

Application : Automated material handling / pick & place operations.

Department : Maintenance

Prepared by : Kalla Naga Chandra kala

Version : 1.0

2. Purpose

To provide the correct operating procedure for the Efort robot system to ensure safe, smooth, and efficient operation during production.

3. Scope

This SOP applies to all operators, technicians, and maintenance personnel working with the Efort robot system.

4. Safety Precautions

Personal Safety:

- Wear PPE: safety shoes, helmet, gloves, mask, hair cover, and safety glasses
- Do not enter inside robot working area during auto mode.
- Never bypass safety interlocks or door sensors.
- Keep hands away from moving parts and robot arm.
- Use Emergency Stop immediately during abnormal conditions

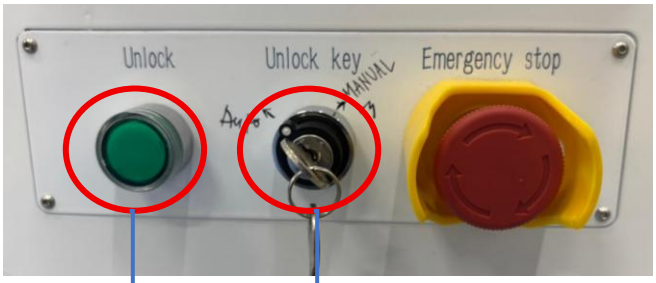
Machine Safety:

- Ensure air pressure and power supply are normal.
- Check robot area is free from obstacles.
- Do not manually move robot arm when servo power is ON.

5. Main Components

- Teach Pendent
- Efort Robot Arm
- Gripper
- Safety Sensors
- Emergency Stop Switch

6. Operating Procedure



2. Press the unlock button.

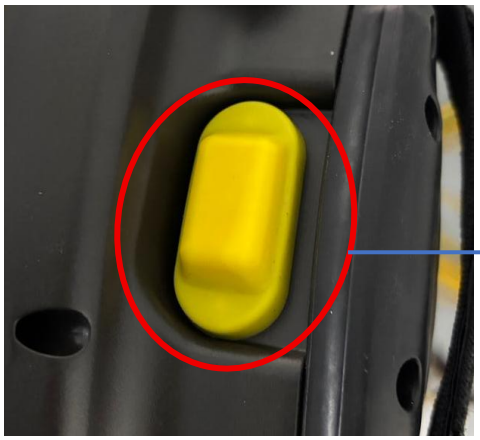
1. Turn the key into manual mode while entering into robot area.

3. Set the Teach Pendant Selector switch in T1 mode while operating in manual mode.

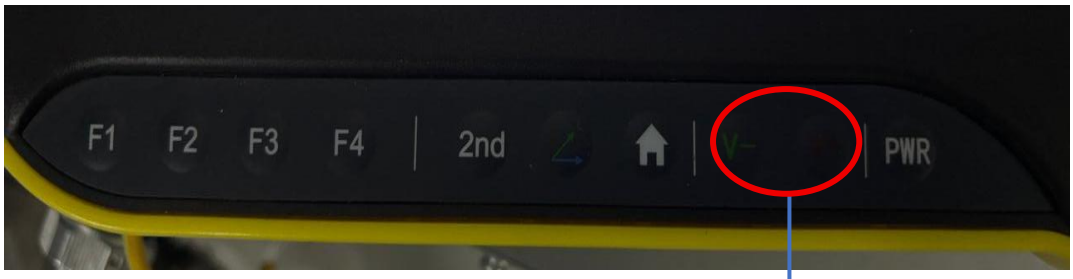




4. Buttons for all robot axis to operate forward and backward.



5. Hold the dead man switch to turn on the servo. It is located at the backside of teach pendent.



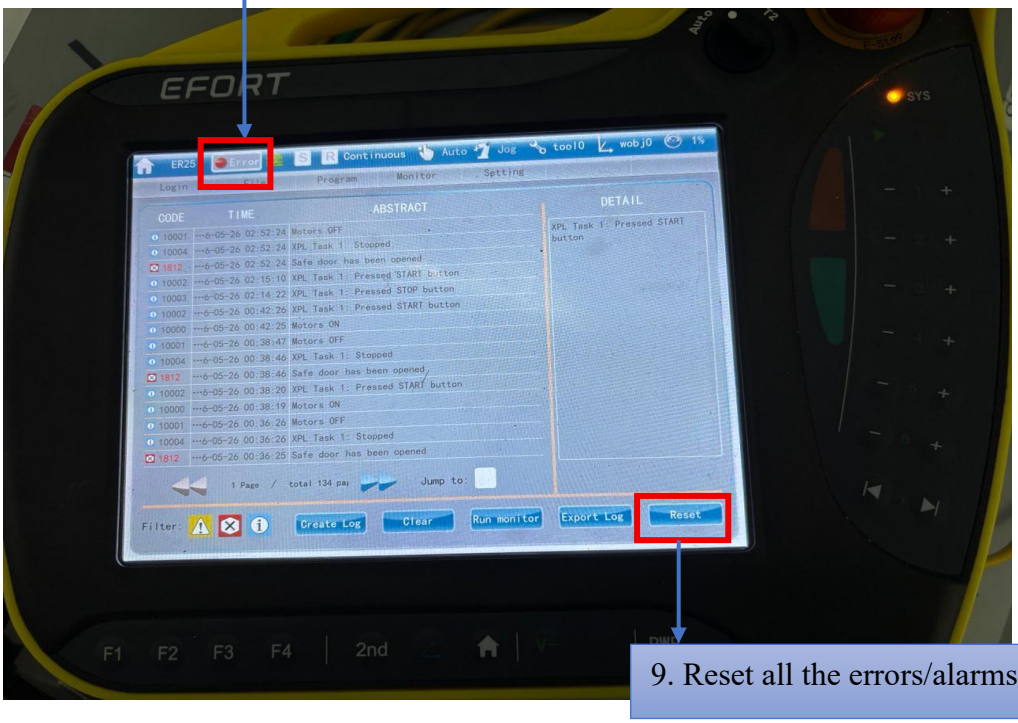
6. Robot Speed controller buttons.



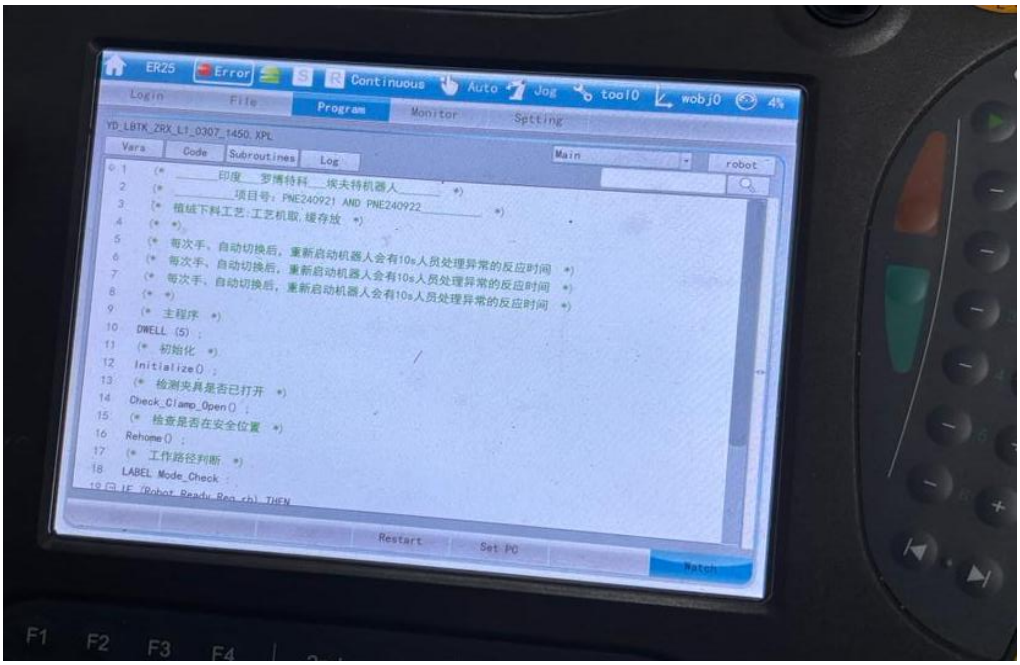
7. LOGIN Page. (Not mandatory everytime)

Enter the credentials and login.

8. Click on ERROR.

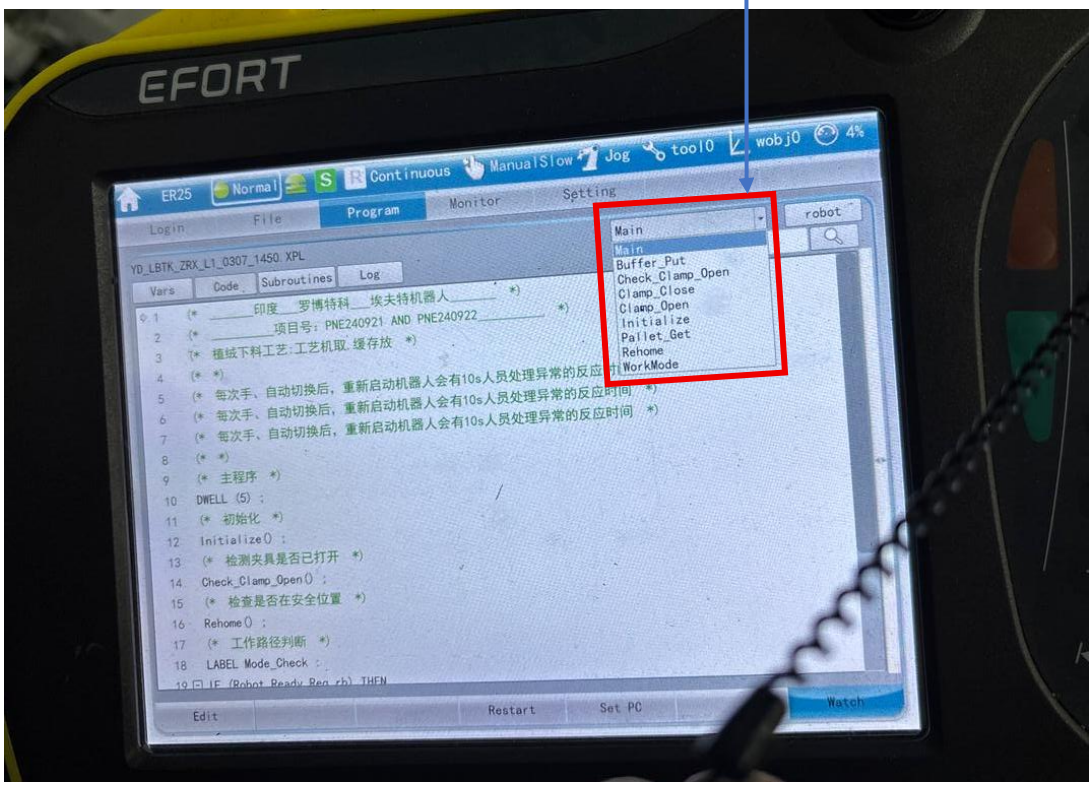


9. Reset all the errors/alarms.



10. The main program of robot operation.

11. Modules of the program for the specific operation.

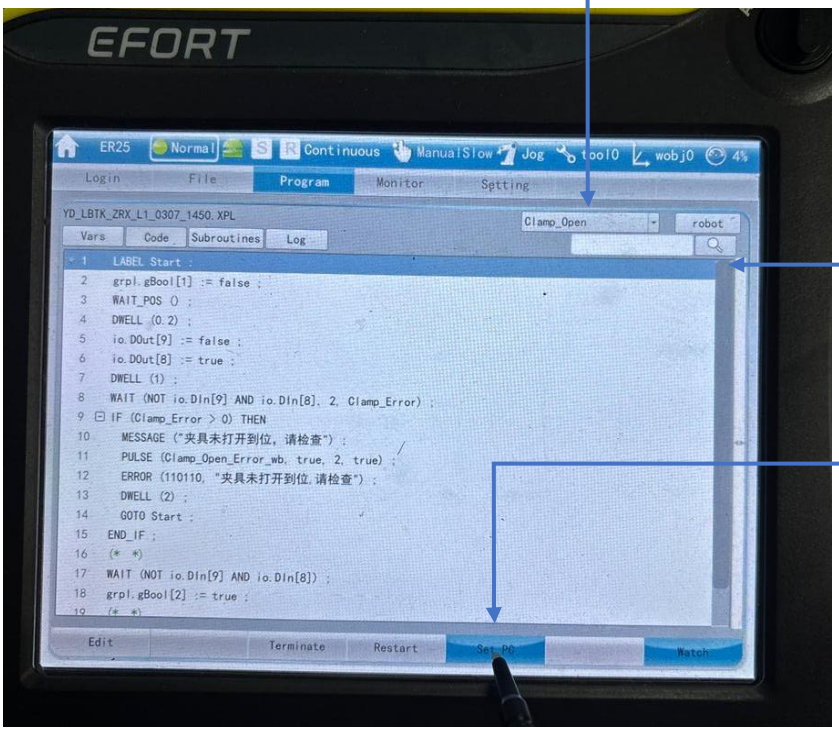


TO CLAMP OPEN:

A. Select the specific module from the list. (Ex: Clamp open)

B. Select the first line or START line of the program

C. Click “SET PC” for operation

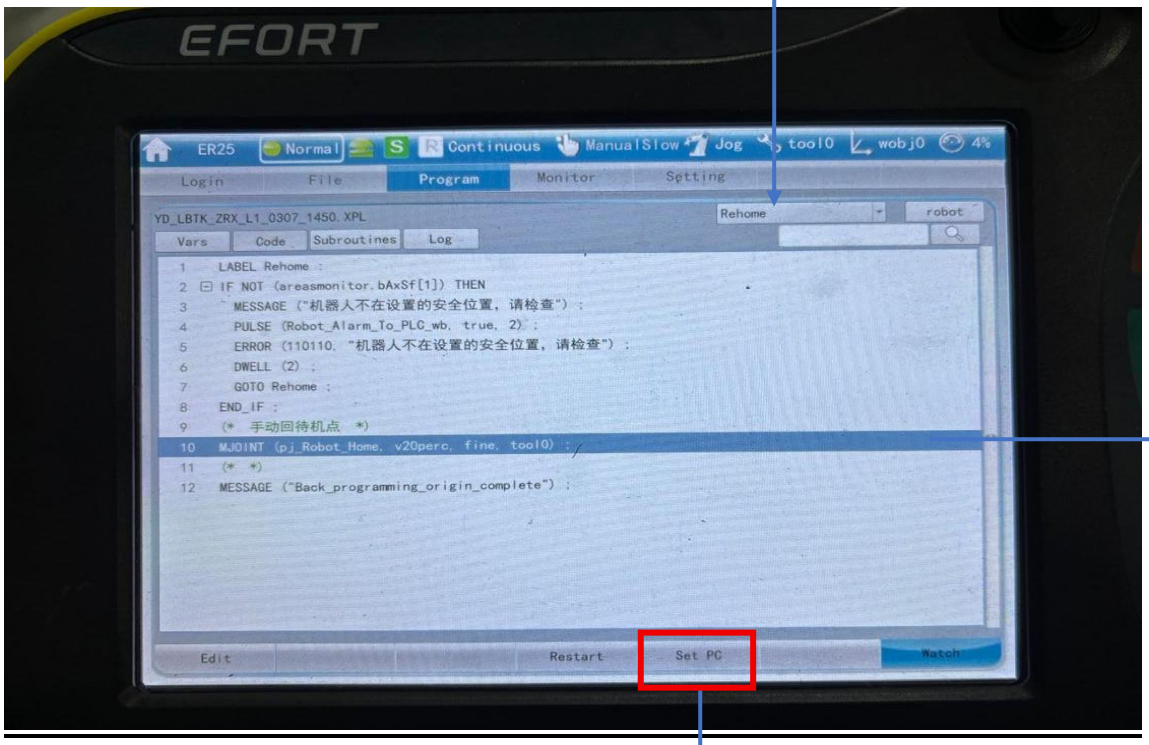




D. Click play button to run the selected operation.

TO REHOMING THE ROBOT:

A. Select the specific module from the list. (Ex: Rehome)



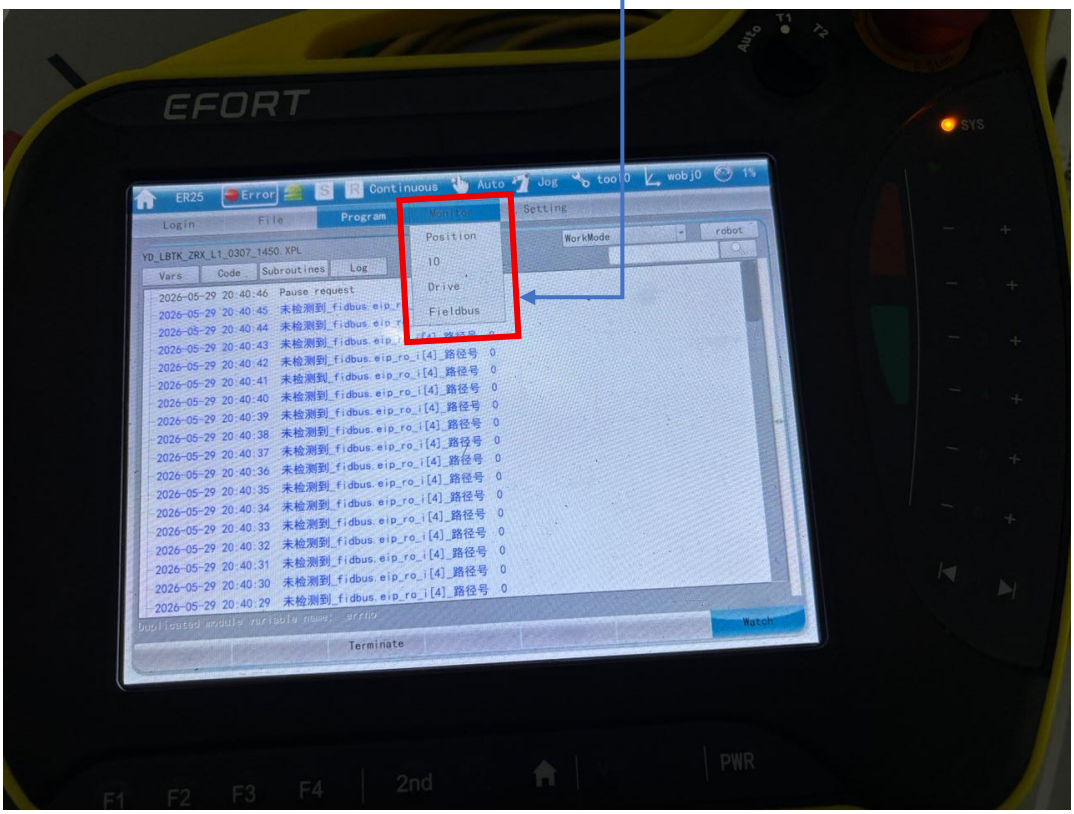
C. Click “SET PC” for operation

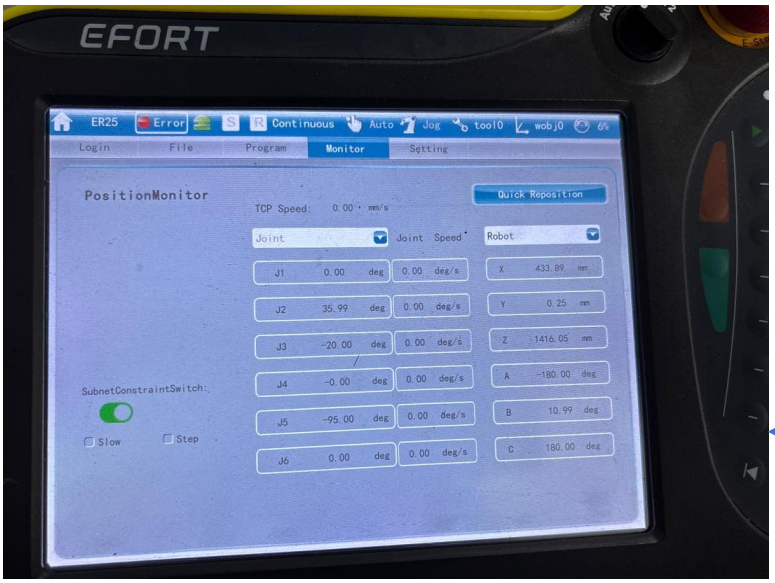
B. Select the run program line.



D. Click play button to run the selected operation.

12. Options for a monitor icon.





13. Angle of all robot axis will be displayed here. Confirm all the axis will be 0 deg after homing.



14. After the manual operations , at the time of exiting the robot place we should acknowledge the safety selector switch.

Trail RUN :

- Remove LOTO.
- Ensure the robot is in the Home position.
- Run the robot in manual mode and verify all axis movements.
- Observe for abnormal noise, vibration or alarms.
- Verify pick-and-place accuracy and alignment.